

Construction of DNA Neural Network Based on Combinatorial Conformational Change

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Abstract

With the integrated development of life sciences and information science, DNA computing is emerging as a crucial research direction to overcome the bottlenecks of traditional computing models. By utilizing DNA for data storage, DNA computing solves complex computational problems through the design and manipulation of interactions between DNA molecules. Leveraging the allosteric properties of DNA molecules, we constructed a programmable molecular component based on a combinatorial allosteric mechanism. By incorporating convolution multiply-accumulate operations and a winner-takes-all strategy, we built a complete DNA neural network, which was then used to successfully perform pattern recognition tasks for 144-bit handwritten Arabic numerals 1-8. For the same recognition task, this molecular component reduces DNA strand complexity by approximately 51% compared to molecular networks constructed with other components. The programmable DNA molecular component is not only applicable to DNA neural network but also suitable for DNA nanorobots and molecular adaptive systems.

CCS Concepts

• **Applied computing** → Life and medical sciences; Computational biology.

Keywords

DNA computing, DNA neural network, DNA strand displacement, pattern recognition, convolution operation

ACM Reference Format:

Xuedong Zheng and Yu Fu. 2025. Construction of DNA Neural Network Based on Combinatorial Conformational Change. In *2025 2nd International Conference on Intelligent Computing and Data Analysis (ICDA 2025)*, August 22–24, 2025, Zhengzhou, China. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3772726.3772818>

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1 Introduction

Beyond its role as the carrier of genetic information storage and transmission in living organisms, DNA possesses inherent properties such as high parallelism, self-assembly capability, and ultra-high data storage density, making it particularly suitable for implementing molecular-level biocomputing. As an emerging interdisciplinary field bridging life sciences and information sciences, DNA computing employs DNA molecules as storage media. By designing base pair arrangements to control DNA hybridization [1] and modulating reaction kinetics through methods like adjusting binding domain lengths [2] [3], it achieves data storage and numerical computation through molecular interactions during biochemical processes, offering a novel approach to overcome limitations of conventional computing paradigms. In 2000, Professor Yurke pioneered the toehold-mediated DNA strand displacement technique [4], realizing molecular-scale binary state switching and establishing a universal platform for constructing DNA circuits and neural networks. Owing to its exceptional specificity, selectivity, and programmability, DNA strand displacement has been extensively investigated and applied in DNA computing domains including logic operations [5] [6], programmable gate arrays [7], biosensing [8], and nanotechnology [9]. Recent advances in programmable DNA circuitry have enabled the successful construction of diverse, scalable molecular systems. Particularly, DNA allosteric mechanisms confer highly flexible dynamic regulation capabilities, allowing molecular systems to adaptively respond to environmental stimuli, thereby enhancing system intelligence and adaptability while enabling parallel computation. Current research on DNA-based regulatory mechanisms spans multiple domains including programmable allosteric control [10] [11], cooperative conformational regulation [12], dynamic allostery [13] [14], and intramolecular conformational dynamics [15]. Furthermore, DNA neural networks [16], as an emerging paradigm integrating neuromorphic and biological computing, demonstrate unique advantages in complex tasks such as combinatorial optimization and pattern recognition [17] [18], offering innovative approaches for biocomputing and intelligent system development with promising AI applications.

To reduce DNA strand complexity and encoding difficulty in DNA neural networks, this study develops a novel DNA neuron based on combinatorial allosteric regulation [19] and constructs a DNA neural network incorporating convolution operations and winner-take-all strategy. The network achieves parallel multiply-accumulate (MAC) operations and handwritten digit recognition. Chemical reaction networks (CRNs) simulations demonstrate that this allostery-based DNA convolutional network exhibits enhanced programmability while significantly reducing strand complexity



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ACM ISBN 979-8-4007-2020-8/2025/08
<https://doi.org/10.1145/3772726.3772818>

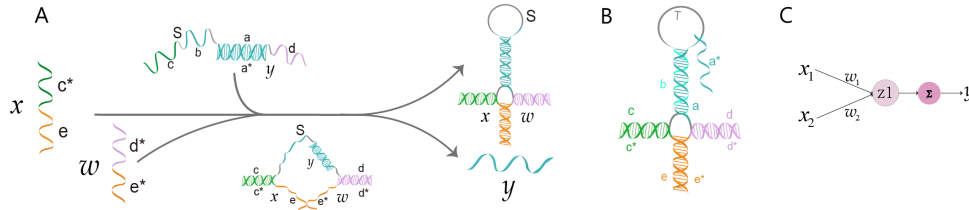


Figure 1: (A) Combinatorial allosteric mechanism. (B) Allosteric regulatory domains. (C) Neuron structure.

[20], showing potential for large-scale data processing and parallel computing applications.

2 Results and discussions

2.1 Construction of DNA neurons based on combinatorial allosteric mechanism

From Figure 1A, by pre-designing DNA domains, the input strand x and w trigger a conformational change in DNA complex Sy . As strand y dissociates, a stable allosteric product Swx and the output strand y are formed. The molecular logic can be simply described by the equation: $x + w + Sy \rightarrow \{k\}Swx + y$. Furthermore, the allosteric product (Figure 1B) exhibits various regulatory domains. Through modular encoding of the structure, the sequence design of the four “arms” enables the regulation of data storage and computational relationships between two input strands and one output strand. For example, n x -strands can react with the same w -strand to output i y -strands, or n x -strands can react with m w -strands to output the same y -strand, which means the product achieves both recognition of input and control over specified output signal pathways, demonstrating high programmability.

Based on the combinatorial allosteric mechanism, we can analogize the interactions between molecules to neuronal signal transmission, where strand x serves as the neuron’s input, strand w acts as the weight, the allosteric process represents signal propagation, and strand y functions as the neuron’s output. This conceptual framework enables the construction of a DNA-based neuron. For the neuron in Figure 1C, the structure and function of the neuron can be encoded using DNA based on the combinatorial allosteric mechanism. By designing substrates with identical output regulatory domains, both multiplication and summation operations can be achieved through a single combinatorial allosteric reaction. Due to the structural modularity of allosteric product, we can selectively regulate one arm to enable substrate sharing, ensuring one neuron corresponds to one substrate. Specifically, the product x_1w_1 is computed through a combinatorial allosteric reaction triggered by x_1 , w_1 , and substrate Sy ; the product x_2w_2 is computed via an analogous reaction involving x_2 , w_2 , and the same substrate Sy . Both biochemical reactions generate the same output y , achieving summation while significantly reducing the interaction complexity.

2.2 Construction of DNA neural network

Based on the above DNA neurons, we propose a DNA neural network construction strategy with MAC functionality by incorporating convolution operations. From Figure 2A, a 2×2 convolution operation is converted into molecular implementation. Using the

input pattern x_1-x_4 , and a 1×2 convolution kernel w_1-w_2 , sliding calculation is performed with a unit step (stride=1) to output y_1 and y_2 such that $x_1w_1 + x_3w_2 = y_1$ and $x_2w_1 + x_4w_2 = y_2$.

In the molecular implementation, all parameters are encoded as distinct DNA strands: the input are encoded as four unique strands (x_1-x_4), while the weight parameters are encoded as two separate strands (w_1 and w_2). Given the weight-sharing characteristic in convolution operations, we leverage the four-arm regulatory feature of the combinatorial allosteric components, enabling different input strands to undergo allosteric reactions with the same type of weight strand and form distinct output. From Figure 2B, both the strands x_1 and x_2 react with the weight strand w_1 , thereby achieving weight-sharing. Additionally, to reduce the number of molecular species, the strand x_1 and x_3 undergoes a combinatorial allosteric reaction with the substrate Sy_1 to produce y_1 . In this way, the substrate Sy_1 functions as a summation gate, triggering two distinct molecular reactions that both yield y_1 , thereby implementing a summation. This design also stems from computations within the same DNA neuron, ensuring that operations belonging to the same neuron share the same summation gate. To handle negative weights, an additional subtraction gate is introduced. As shown in Figures 2C and 2D, negative and positive weights are encoded and represented by w_1^- and w_2^+ , respectively. The positive summation gates are Sy_1 and Sy_2 , while the negative summation gates also share Sy_1^- and Sy_2^- . The subtraction gate is denoted as Sa . The input first undergoes MAC, generating corresponding output representing positive-weight sums y^+ and negative-weight sums y^- . These upstream reaction products, y^+ and y^- , then serve as inputs to the downstream subtraction module, where they are stoichiometrically consumed by subtraction gate Sa . Upon reaction completion, the remaining concentration of y^+ is designated as the molecular network’s output, while all other products are treated as waste. This outputs selection strategy effectively implements the functionality of a ReLU activation function. Notably, employing the same encoding regulation mechanism, the subtraction gate is also designed to share substrate strands, thereby reducing network complexity. The calculation process can be summarized as the equations: $-x_1w_1 + x_3w_2 = y_1$, $-x_2w_1 + x_4w_2 = y_2$, $Y_1 = Relu(0, y_1)$ and $Y_2 = Relu(0, y_2)$.

Figure 3 presents the CRNs simulation results for the 2×2 molecular convolution operation implemented in Visual DSD, where the input patterns 1/0 represent the presence/absence of DNA strands. Six typical combinations (0001, 1000, 1001, 1110, 0111 and 1111) are selected to conduct simulation validations for both positive and negative weight tasks. The simulated results demonstrate agreement

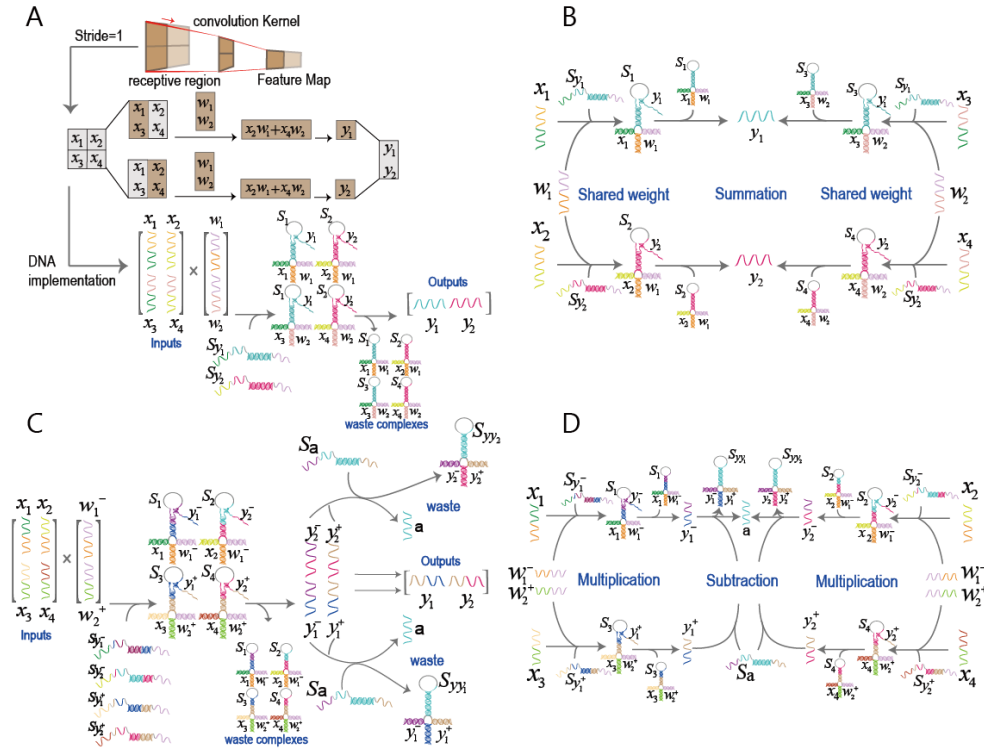


Figure 2: (A) Schematic diagram of molecular implementation of convolution operation. (B) Molecular reaction strategy for weight sharing. (C) Molecular implementation for handling negative weights. (D) Molecular strategy for shared negative weights.

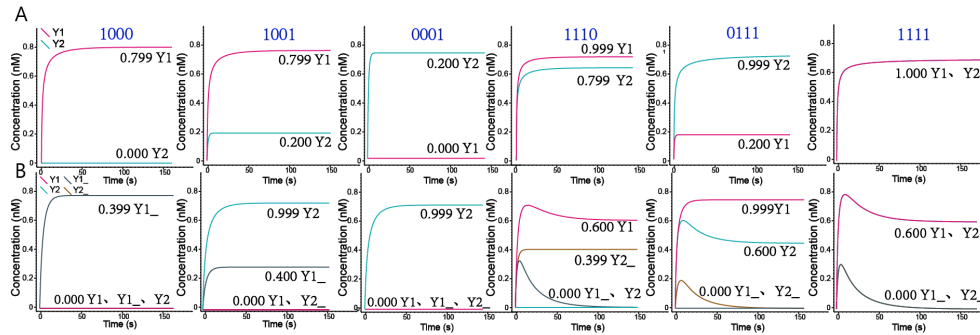


Figure 3: (A) Concentration changes for positive weights. The initial concentrations of reactants were 1nM, the relative concentrations of w_1 and w_2 were 0.8 and 0.2, and the rate constant was 1. (B) Concentration changes for negative weights. The initial concentrations of reactants were 2nM, the relative concentrations of w_1^- and w_2^+ were 0.4 and 1, and the rate constants for multiplicative summation and subtraction operations were 0.015 and 0.02.

with theoretical predictions, thereby fully validating the scalability of the network strategy.

The architecture of DNA neural network is presented in Figure 4A, where each neuron Z_i processes a local receptive field from the input layer and computes feature values (i.e., a subset of input patterns $\{x_n\}$). Different receptive fields share a common set of weight values $\{w_m\}$. After undergoing multiply-accumulate operations,

subtraction computations, and ReLU activation processing, the maximum concentration value of the output strands is selected as the final network output. To achieve the winner-takes-all strategy, an annihilation catalysis module is further incorporated as shown in Figure 4B, where the initial concentrations satisfy the constraint $[y_i] < [S_j p_i] < [F] < \sum [y_i]$. Therefore, the limited catalyst F triggers kinetic competition among species y_i : the higher the initial $[y_i]$, the higher $[j_i]$ generated by its annihilation reaction, allowing

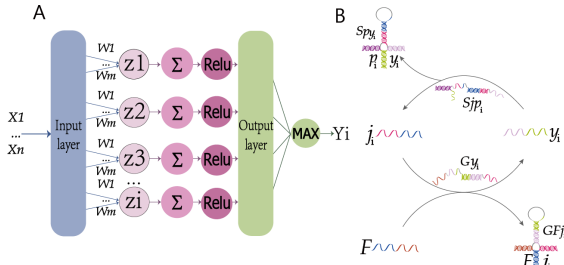


Figure 4: (A) DNA-based neural network architecture. (B) Molecular reaction pathway in catalytic annihilation modules.

it to compete for more catalyst and trigger catalytic reactions that restore $[y_i]$ while consuming its annihilation substrate. However, since the initial catalyst concentration is insufficient to sustain all y_i catalytic reactions, the y_i with the highest initial concentration preferentially depletes all of its annihilation substrate and survives, while other species are eventually eliminated due to insufficient catalysis and consumption by the annihilation substrate. The reaction logic of this DNA neural network can be uniformly described by the following CRNs where $n, m, i \in \{1, 2, 3, 4, \dots\}$:

$$\begin{aligned} &\text{MAC Parallel Computing} \\ &\begin{cases} x_n + w_m + Sy_i \rightarrow \{k_n\} S_n w_m x_n + y_i \\ x_n + w_m + Sy_i^- \rightarrow \{k_n\} S_n w_m x_n + y_i^- \end{cases} \end{aligned} \quad (1)$$

$$\begin{aligned} &\text{Subtraction\&Relu Function :} \\ &y_i + y_i^- + Sa \rightarrow \{f_i\} Sy_i + a \end{aligned} \quad (2)$$

$$\begin{aligned} &\text{Annihilation Matching} \\ &\begin{cases} y_i + Sj_i \rightarrow \{k_i\} j_i + Sp_i y_i \\ j_i + F + Gy_i \rightarrow \{k_i\} y_i + GFj_i \end{cases} \end{aligned} \quad (3)$$

2.3 Network applications for pattern recognition tasks

To validate the feasibility of the DNA neural network framework, we conducted pattern recognition tasks for 144-bit handwritten Arabic numerals 1-8. As shown in Figure 5A, grayscale images of handwritten digits selected from the MNIST open database (<http://yann.lecun.com/exdb/mnist/>) were scaled to a 12×12 grid, with grid pixel values above the threshold set to 1 and the remaining bits set to 0. In the molecular implementation, by sequence design, the 12×12 input data is encoded into DNA strands $x_1 - x_{144}$, while a 6×3 convolution kernel is encoded into DNA strands $w_1 - w_{18}$ with a stride of 6. Each slide corresponds to a neuron computation, specifically the sequential output of $\{y_1 - y_8\}$ and $\{y_1^- - y_8^-\}$ through eight neurons. The corresponding reaction substrates are the positive summation substrates $\{Sy_1 - Sy_8\}$, negative summation substrates $\{Sy_1^- - Sy_8^-\}$, subtraction substrate Sa , annihilation substrates $\{Sj_1 - Sj_8\}$, and catalytic substrates $\{Gy_1 - Gy_8\}$. The weights were obtained through model training using a dataset of 8,000 images (including both training and test sets). As shown in Figure 5B, the model achieved an accuracy of 90% and completed the classification task on the validation set (1,000 samples per category) with the classification results displayed in Figure 5C. Finally,

the trained weights were extracted and normalized to $[-1, 1]$ using max-value normalization.

As shown in Figure 6C, eight pre-identified input patterns were obtained by randomly selecting one sample per category from the validation set. In the input patterns, all input strands corresponding to "1" (gray-scale blocks) were set to an initial concentration of 200nM, with f_i and k_i values of 3×10^{-4} and 3×10^{-3} , respectively. All substrate molecular strands were prepared at 200 nM, while the concentration of weight strands was set as (weight value $\times 100$ nM). The initial concentration of substrate strands in the output layer required adjustment based on the concentration of upper-layer output strands, with the adjustment conforming to the initial concentration requirements. The DNA neural network based on combinatorial allosteric was simulated using Visual DSD to model its CRNs. From Figure 6A, the substrates of the output layer were added after the computation completed. The resulting concentration curves exhibited a declining trend, with the surviving y_i at maximum concentration being identified as the recognition result (output strands $y_1 - y_8$ correspond to target patterns 1-8 respectively).

In the DNA neural network designed for the aforementioned pattern classification task, the molecular strands encoding parameters include: input strands (for patterns with value "1"), a shared set of weight strands, positive-weight substrate strands, negative-weight substrate strands, subtraction substrate strands, annihilation substrate strands, catalytic substrates, and catalytic strands. The corresponding molecular parameter configurations are detailed in Figure 6B. For this recognition task involving 144-bit input patterns across 8 target patterns, alternative molecular strategies require a network size of 346 [20], whereas our approach achieves a network size of just 196. That is, when implementing weight sharing with molecular switch components, weight substrate molecules are needed to activate weight molecules, and the network parameter setting is $n + 5m + (m + 1)be$ (where n denotes the number of bits in a pattern, b denotes the number of groups, m is the number of patterns in each group and e represents the number of bits in a convolution kernel). However, our method can achieve weight sharing only through sequence design, without the need for intermediate steps, and it is directly reflected in the calculation process of product summation. Comparatively, the DNA computing network constructed using combinatorial allosteric molecular components effectively reduces molecular network complexity by approximately 51%.

3 CONCLUSIONS

Although significant progress has been made in DNA neural networks, the complexity and encoding difficulty of molecular reaction networks increase with the scale of computation. How to construct a programmable, low-complexity DNA neural network remains a critical challenge. This paper proposes a construction strategy for DNA neural networks based on a combinatorial allosteric mechanism, which can effectively reduce the complexity of molecular reaction networks. Notably, all signal transmissions in this study are achieved using molecular components based on combinatorial allosteric. Leveraging the unique regulatory mechanism of these components, the flexibility of DNA encoding is enhanced while maintaining programmability and simplicity. The relevant

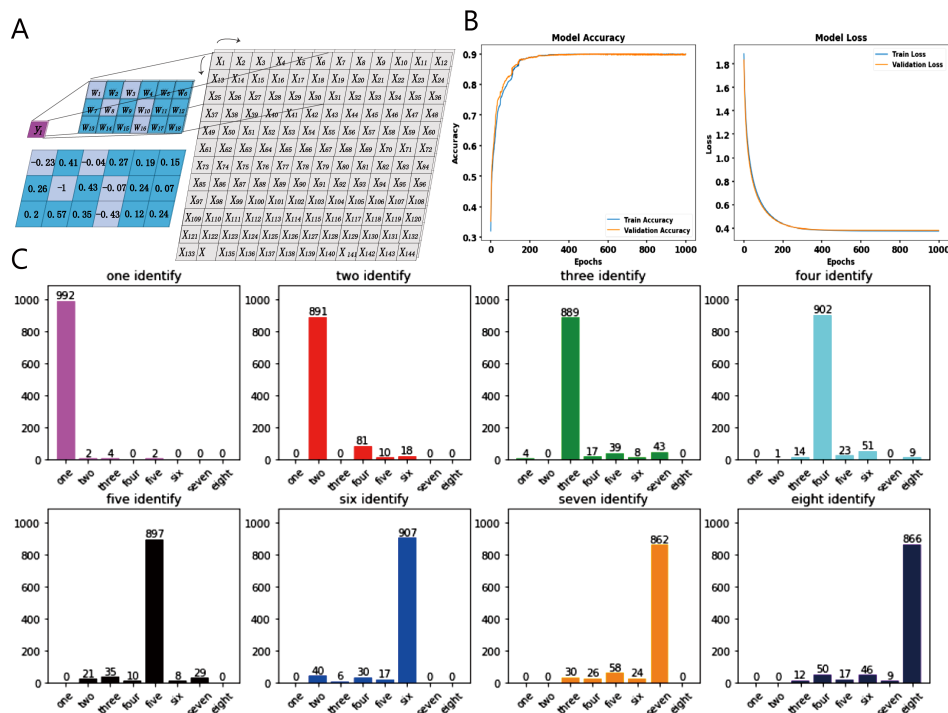


Figure 5: (A) Diagram of computational process and weights. (B) Model training results. (C) Validation results of pattern recognition.

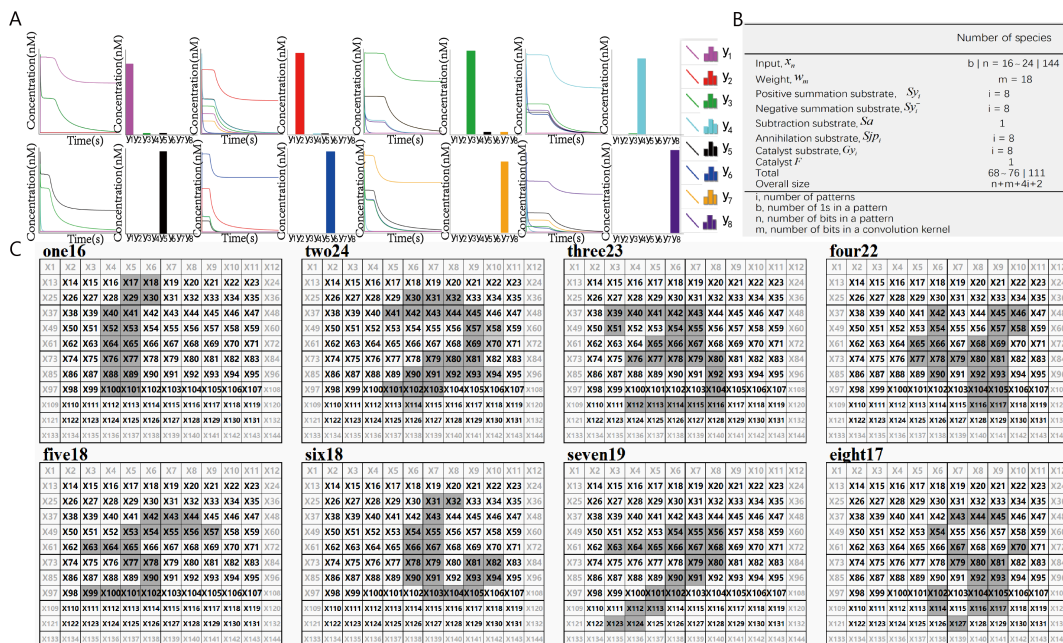


Figure 6: (A) Pattern Recognition Results of Molecular Networks. (B) Parameter Settings and Network Scale of Molecular Networks. (C) Handwritten Arabic numerals "1-8" pre-identified input patterns and the count of 1s contained.

results can be applied to fields such as DNA machines and DNA nanorobots.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 61972266) and the Major Open Project of Key Laboratory for Advanced Design and Intelligent Computing of the Ministry of Education (No. ADIC2023ZD002).

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